Lab 7: Null Space and the Rank-Nullity Theorem

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, October 21st, 2026

Name: _____

uniqname: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Recap: Rank, Column Space, and Null Space

- For an $n \times d$ matrix A, colsp(A) is a subspace of $\mathbb{R}^n$; it is the span of A's columns.
- rank$(A) =$ dim(colsp(A)).
- nullsp$(A) = \{\vec{x} \in \mathbb{R}^d : A\vec{x} = \vec{0}\}$.
- The **Rank-Nullity Theorem** states that

$$\text{rank}(A) + \dim(\text{nullsp}(A)) = d$$

- rank$(A^T) =$ rank(A).

Activity 1: Null Space of a Matrix with Linearly Independent Columns

Let $A = \begin{bmatrix} 3 & 0 \\ 0 & 4 \\ 1 & 0 \end{bmatrix}$. What is nullsp$(A)$?

Activity 2: Null Space from Column Relationships

Let $X = \begin{bmatrix} 1 & 2 & -1 & 3 & 4 & 4 \\ 2 & 5 & -2 & 7 & 11 & 10 \\ 4 & 8 & -4 & 12 & 16 & 16 \end{bmatrix}$.

Note that X's first two columns form a basis for its column space; verify this before proceeding.

a) Fill in the blanks: nullsp(X) is a ____-dimensional subspace of ____.

b) Find a basis for nullsp(X). *Hint: You should be able to answer this without solving equations.*

Activity 3: Four Spaces Warm-Up

Let A be a 7×9 matrix with rank 5. For each space below, state its dimension and the space $\mathbb{R}^k$ of which it is a subspace.

Space	Dimension	Subspace of $\mathbb{R}^k$ (state k)
colsp(A)		
colsp(A^T)		
nullsp(A)		
nullsp(A^T)		

Activity 4: Constructing a Matrix

Find a matrix A such that

$$\begin{bmatrix}3\\1\end{bmatrix} \in \text{colsp}(A) \qquad \text{and} \qquad \begin{bmatrix}1\\3\end{bmatrix} \in \text{nullsp}(A).$$

Activity 5: The Systems of Equations View

Let A be an $n \times d$ matrix of rank r, and suppose there exist vectors $\vec{b} \in \mathbb{R}^n$ such that

$$A\vec{x} = \vec{b}$$

does not have a solution (meaning no $\vec{x}$ makes $A\vec{x} = \vec{b}$).

a) What are all inequalities ($<$ or $\leq$) that must be true between n, d, and r?

b) How do you know that $A^T\vec{y} = \vec{0}$ has solutions other than $\vec{y} = \vec{0}$?

Activity 6: Columns in a Null Space

Let A and B be non-zero 6×6 matrices. Suppose $\text{rank}(A) = 4$, and that every column of B is in $\text{nullsp}(A)$.

a) Show that every row of A is orthogonal to every column of B.

b) Find AB and $\text{rank}(AB)$. Justify your answers.

c) What are the possible values of $\text{rank}(B)$? Explain why these are the only possibilities, and why each one is possible.

Activity 7: Can the Column Space Equal the Null Space?

In each part, determine whether it's possible to find a matrix whose column space and null space are the same. If it is, give an example and show that it works. If not, explain why not.

a) Can you find a 2×2 matrix A such that $\text{colsp}(A) = \text{nullsp}(A)$?

b) Can you find a 3×3 matrix A such that $\text{colsp}(A) = \text{nullsp}(A)$?